

## Chemistry:-

MCQs :-

1. A

2. B

3. A

4. D

5. A

6. B

7. C

8. D

9. A

10. B

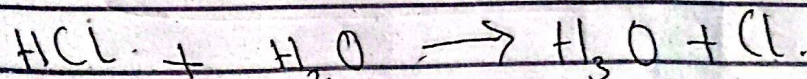
Q. 2

(i) Brønsted Lowry acid :-  
An acid is proton donor  
that means an acid donates  
a ~~acid~~ protons.

Brønsted Lowry base :-

A base is proton acceptor.  
that means a base accepts  
a ~~acid~~ protons.

(ii)



~~Explanation :-~~

$\text{H}_3\text{O}^+$  } Brønsted acid }

Reason - HCl is an acid because  
it donates protons. whereas

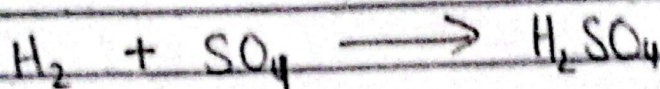
H<sub>2</sub>O is base and that's why  
H<sub>2</sub>O is acids because HCl accepts  
donates protons and H<sub>2</sub>O  
accepts a protons.

Q.3

(i) HNO<sub>3</sub>

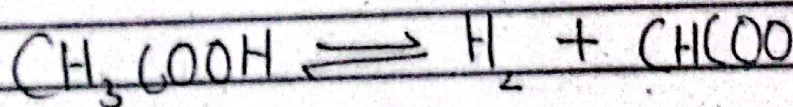
$H + NO_3 \rightarrow HNO_3$   
{ it is a strong acid because it }  
ionizes completely }

(ii) H<sub>2</sub>SO<sub>4</sub>



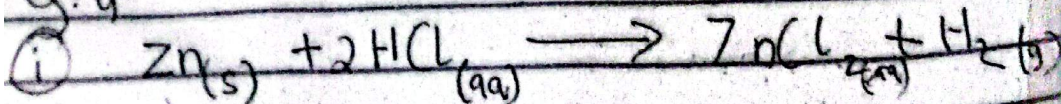
{ it is a strong acid because it }  
ionizes completely }

(iii) CH<sub>3</sub>COOH

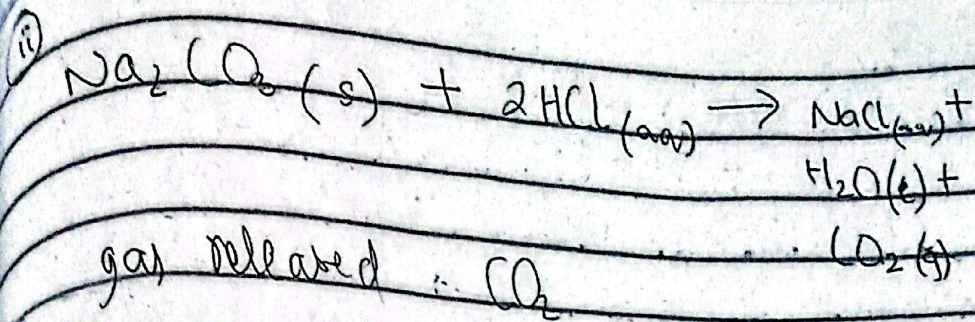
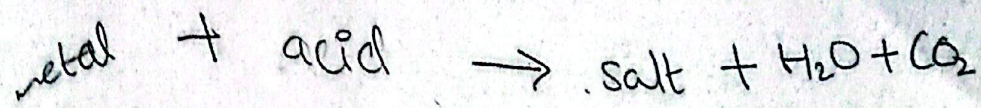


{ it is a weak acid because }  
{ it do not ionizes completely }

Q.4



gas released = hydrogen gas



Q:5

(1) All alkalis are bases becz alkalis are bases that dissolve in water and produce  $\text{OH}^-$  ions whereas Not all bases are alkalis becz some bases do not dissolve in water.

Example :-

Bases that is an Alkali :-  
 $\text{NaOH}$

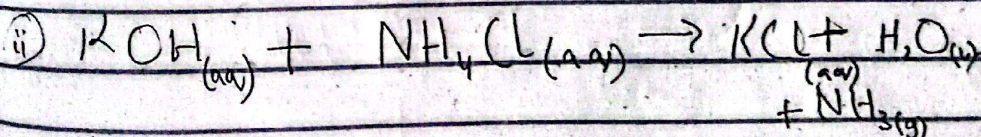
Base that is not Alkali :-  
 $\text{CuO}$

Q:6

(i)

1. Bases change the colour of red litmus paper to blue.

2. Base neutralises acids to form salt and water.



Q.7

(i) molar mass:-  
 $23 + 16 + 1$   
 $= 40 \text{ g/mol}$

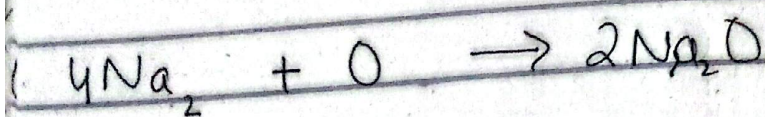
(ii) mass = m.m  $\times$  mole  
 $40 \times 0.2$   
 $8$

(iii) formula units :- mole  $\times N_A$   
 $= 0.2 \times 6.022 \times 10^{23}$   
 $= 1.2044 \times 10^{23}$

Q.8

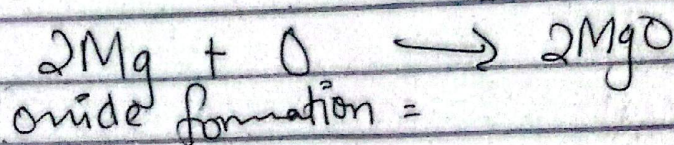
(i)

(a) Sodium:-

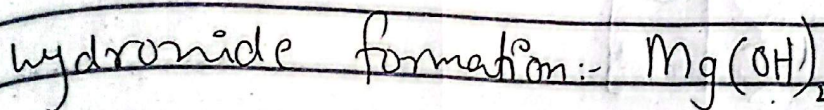
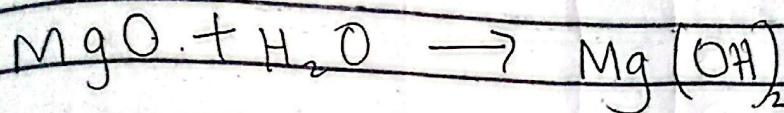
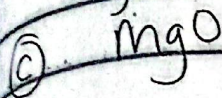
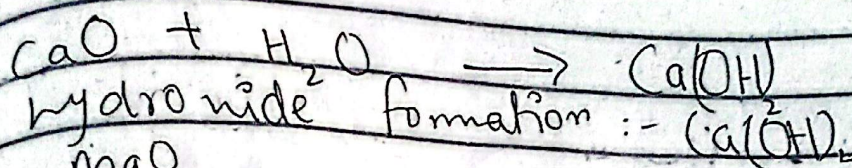
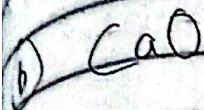
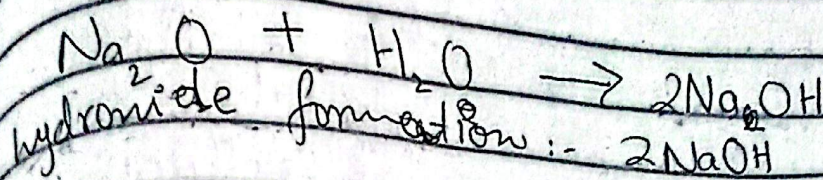
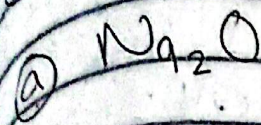


oxide formed :-  $2\text{Na}_2\text{O}$

(b) magnesium:-

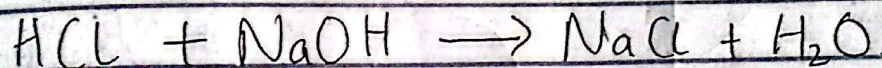
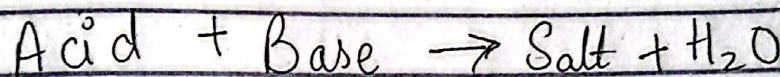


(ii)



(iii)

general equation :-

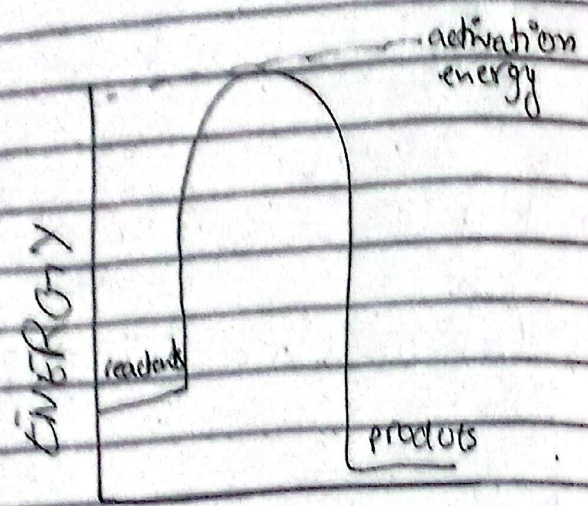


always formed :- Salt + water.

Q.1

i) Activation energy  
The minimum amount of energy that in addition to the average kinetic energy where particles must have an effective collisions called activation energy.

(ii)



(iii)

